

CCNA 200-301 - Chapter 7 Summary

Static IP: Manual configuration (IP, Mask, Gateway, DNS).
Used for Servers, Printers, Routers.

DHCP: Automatic IP assignment.

DORA: Discover (Broadcast) → Offer → Request → ACK.

Discover packet:

Source IP: 0.0.0.0

Destination IP: 255.255.255.255

Destination MAC: FF:FF:FF:FF:FF:FF

Alternate IP: Backup configuration if DHCP is unavailable.

APIPA

Range: 169.254.x.x

Mask: 255.255.0.0

Appears when DHCP fails. No Internet.

Commands

`ipconfig /release`

`ipconfig /renew`

IP Classes

A: 1-126 → 255.0.0.0

B: 128-191 → 255.255.0.0

C: 192-223 → 255.255.255.0

Loopback: 127.0.0.1

Private Ranges

10.0.0.0/8

172.16.0.0 - 172.31.255.255

192.168.0.0/16

ARP

Unknown MAC → Broadcast ARP → Target replies with MAC.

Cables

Max UTP length: 100m. After that use a Switch/Repeater.

Straight: Different devices.

Crossover: Same devices.

Rollover: Console.